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BPSC SPECIAL NOTES

TRE 4.0 Computer Science

Computer Hardware

TRE 4.0 CS का पहला topic: CPU, Memory, Bus

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Computer System और CPU

(1) Block diagram

- Computer = Input, Process, Output, Store (IPO cycle)
  → CPU = CU + ALU + Registers
- Digital computer के 3 main parts: memory, CPU, I/O (TRE 2.0)
- Brain of computer = CPU (TRE 2.0)

(2) Architecture

- Von Neumann: program और data एक ही memory में (stored program)
- Harvard: program और data की अलग memory और bus
- Father of Computer = Charles Babbage (TRE 1.0)

(3) CPU के हिस्से

- ALU: arithmetic और logic operations
- CU: instructions decode करके control signals बनाता है (TRE 1.0, दो बार पूछा)
- Chips का primary element = Silicon (TRE 1.0)

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Evolution और Generations

① Timeline

- Pascaline (Pascal, 1642)
- Analytical Engine (Babbage, 1834)
- Transistor (1947, Bell Labs)
- Moore's Law: हर 2 साल में transistors double

② Generations

- 1st: Vacuum tube
- 2nd: Transistor
- 3rd: IC
- 4th: VLSI / Microprocessor
- 5th: AI based

③ Devices

- Video games के लिए = Joystick (TRE 2.0)
- Laser होता है CD/DVD drive में (TRE 2.0)

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Memory के प्रकार

① Units

- Bit (0/1), Nibble = 4 bits, Byte = 8 bits

② Primary memory

- RAM: volatile, read/write, fixed time access (TRE 2.0)
- ROM: non-volatile; BIOS और boot loader
- Cache: CPU और RAM के बीच तेज़ छोटी memory; मिला तो hit, नहीं तो miss

③ Secondary memory

- HDD, SSD, CD/DVD, pen drive
- CPU इसे directly access नहीं करता

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SRAM, DRAM और ROM types

① SRAM vs DRAM

- SRAM: flip-flop, refresh नहीं चाहिए, तेज़, महँगी, Cache में
- DRAM: capacitor, refresh चाहिए, सस्ती, *Main memory* में

→ दोनों volatile

② ROM types

- MROM: factory में ही data
- PROM: user एक बार program करे
- EPROM: UV light से erase
- EEPROM: बिजली से erase; Flash इसी का block-erase रूप

③ TRE 3.0

- PROM, EPROM, MROM तीनों ROM के types (TRE 3.0)

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Memory Hierarchy और Buses

(1) Hierarchy

- ऊपर से नीचे: speed और cost per bit घटती, capacity बढ़ती
- Cost per bit minimum = magnetic tape (TRE 2.0)
- Cache best for small loops; Interleaved best for large sequential code

(2) System bus

- Data bus: bidirectional
- Address bus: unidirectional, CPU से memory
- Control bus: read/write signals

(3) Formula

- n address lines = 2^n locations
- 8085: 16 lines = 64 KB; 8086: 20 lines = 1 MB
- FFFF last location = 64K (TRE 2.0)

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याद रखने की ट्रिक

★ Generations: VTIVA = Vacuum, Transistor, IC, VLSI, AI

★ ROM types: M-P-E-E = Mask, PROM, EPROM, EEPROM (UV वाला E पहले)

★ SRAM = Speed + cache; DRAM = Dump (refresh) + main memory

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अब Test करो: 10 PYQ

1. Interrupts which are initiated by an instruction are (UGC NET Jun 2012)

(A) Internal
(B) External
(C) Hardware
(D) Software

2. A complete microcomputer system consists of (UGC NET Jun 2012)

(A) Microprocessor
(B) Memory
(C) Peripheral equipment
(D) All of the above

3. Pipelining strategy is called implement (UGC NET Jun 2012)

(A) instruction execution
(B) instruction prefetch
(C) instruction decoding
(D) instruction manipulation

4. The concept of pipelining is most effective in improving performance if the tasks being performed in different stages: (UGC NET Jun 2012)

(A) अलग-अलग समय
(B) लगभग समान समय
(C) अंतर समान रहे
(D) अंतर अलग-अलग

5. In which addressing mode, the effective address of the operand is generated by adding a constant value to the contents of register ? (UGC NET Dec 2012)

(A) Absolute
(B) Indirect
(C) Immediate
(D) Index

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Test जारी: सवाल 6-10

6. One of the main features that distinguish microprocessor from micro-computers is (UGC NET Jun 2014)

(A) words बड़े होते हैं
(B) words छोटे होते हैं
(C) I/O devices नहीं
(D) None of the above

7. Which one of the following is not an addressing mode ? (UGC NET Jun 2013)

(A) Register indirect
(B) Autoincrement
(C) Relative indexed
(D) Immediate operand

8. Which is not a typical program control instruction ? (UGC NET Jun 2013)

(A) BR
(B) JMP
(C) SHL
(D) TST

9. Interrupt which arises from illegal or erroneous use of an instruction or data is (UGC NET Jun 2013)

(A) Software interrupt
(B) Internal interrupt
(C) External interrupt
(D) All of the above

10. A given memory chip has 14 address pins and 8 data pins. It has the following number of locations. (UGC NET Sep 2013)

(A) 2^8
(B) 2^14
(C) 2^6
(D) 2^12

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कल का topic: Number Systems